The Topological Invariance Principle: Dark Energy as a Geometric Constraint of Spatial Flatness in Einstein-Cartan Cosmology

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In the standard cosmological model, the strict spatial flatness of the observable Universe ($\Omega_K \approx 0$) is treated as an observational constraint requiring fine-tuned initial conditions, while late-time accelerated expansion is attributed to a phenomenological cosmological constant (Λ). In this Letter, we propose an alternative conceptual framework based on Einstein-Cartan gravity, governed by what we term the *Topological Invariance Principle* (TIP). We postulate that global spatial flatness ($\Omega_{total} \equiv 1$) is not a parametric coincidence, but a fundamental constant and a dynamic gauge law of spacetime. It is shown that to preserve this topological invariance against the emergence of material curvature, the geometric coupling induces a spacetime reaction: any primordial fractional spin density (Ω_{spin}) injected prior to recombination algebraically enforces a strictly compensatory torsion debt (Ω_τ), such that $\Omega_\tau^{(initial)} = -\Omega_{spin}^{(peak)}$. We demonstrate that Dark Energy naturally emerges from this formalism: not as an exotic substance, but as the residual and dynamic manifestation of this frozen geometric debt. As cosmic expansion dilutes matter, the fractional torsion density mechanically increases to maintain topological equilibrium, acting at leading order as an effective cosmological constant. This gauge model, whose numerical foundations and implications for dark matter are developed in companion papers [1, 2], provides a natural geometric explanation for cosmic acceleration, consistent with recent observations [3, 5].

INTRODUCTION

Precision cosmology, largely driven by Cosmic Microwave Background (CMB) measurements from the Planck satellite [3], dictates that the observable Universe is remarkably flat, with the curvature parameter $\Omega_K = 0.0007 \pm 0.0019$. Within the standard ΛCDM framework, this flatness is a highly unstable equilibrium requiring the ad hoc mechanism of primordial inflation to solve the fine-tuning problem. Furthermore, the accelerated expansion of the Universe is attributed to a cosmological constant (Λ), representing approximately 70% of the total energy density today, yet lacking a fundamental microscopic origin.

In this Letter, we introduce a paradigm shift regarding the nature of spatial flatness and cosmic acceleration. Extending general relativity to the Einstein-Cartan (EC) framework—where spacetime torsion is algebraically coupled to the intrinsic spin of fermions [4]—we propose that spatial flatness is not an accident of initial conditions, but a fundamental invariance law actively maintained by the geometry of spacetime.

The key phenomenological calibration—the sound-horizon reduction and H_0–S_8 joint resolution—is established in the companion paper Foundation I [1]; the microscopic dark sector and baryogenesis mechanism are developed in Foundation II [2].

THE TOPOLOGICAL INVARIANCE PRINCIPLE

We formulate the *Topological Invariance Principle* (TIP) as follows: *Global spatial flatness ($\Omega_{total} \equiv 1$) is an absolute and inviolable geometric equilibrium state, not a contingent initial condition. Any dynamical perturbation of this metric via the injection of a chiral curvature source—specifically, a primordial fractional spin density (Ω_{spin}) prior to recombination—instantly generates a compensatory topological reaction from the spacetime continuum (the macroscopic torsion density, Ω_τ), whose scale-dependent structure is detailed in the companion papers [1, 2].*

Under this principle, the standard Friedmann equation for fractional densities is extended to include the spin-torsion pair:

$$\Omega_m + \Omega_r + (\Omega_{spin} + \Omega_\tau) \equiv 1 \tag{1}$$

where Ω_m encompasses both baryonic and topological dark matter [2], and Ω_r is the radiation density. The parenthetical grouping makes explicit that Ω_{spin} and Ω_τ form a *conjugate pair*: they are not independent components but two faces of the same geometric coupling.

To preserve topological invariance against the massive injection of chiral spin during the early radiation era, the emergent spin density must be perfectly neutralized by the geometric background at the moment of peak injection:

$$\Omega_\tau^{(initial)} = -\Omega_{spin}^{(peak)} \tag{2}$$

This initial condition fixes the magnitude of the topological debt created at the onset of the spin phase. During the active spin epoch, the continuous compensation constraint holds instantaneously:

$$\Omega_{spin}(a) + \Omega_\tau(a) = 0 \tag{3}$$

Torsion is thus fundamentally defined as the geometric counter-force required to maintain a strictly flat Universe in the presence of extreme fermion spin densities. The fate of this debt once the spin fluid dilutes is the subject of the next section.

DARK ENERGY AS A FROZEN GEOMETRIC DEBT

This strict geometric reaction provides a natural, first-principles resolution to the nature of Dark Energy. In the companion paper *Foundation I* [1], we demonstrated that resolving the H_0 tension and lowering the comoving sound horizon to $r_s^{\mathrm{ECF}} = 135.8\,\mathrm{Mpc}$ requires a transient fractional spin density peaking at $\Omega_{spin}^{(peak)} \approx +0.093$ at a redshift $z \approx 7500$ prior to recombination.

According to the TIP (Eq. 2), this transient primordial spin phase obligatorily generated a global topological debt:

$$\Omega_\tau^{(initial)} \approx -0.093 \tag{4}$$

within the spacetime manifold. Crucially, while the kinetic spin fluid dilutes rapidly as a^{-6} due to cosmic expansion, the corresponding macroscopic torsion field (the geometric debt) undergoes a phase transition, "freezing" into the elastic structure of the vacuum once the spin source vanishes. Its absolute energy density (ρ_τ) then becomes constant; the thermodynamic and field-theoretic derivation of this freezing mechanism is developed in Foundation II [2]. At leading phenomenological order, this frozen torsion acts as an effective cosmological constant ($w_\tau = -1$). The full dynamical equation of state—an inherently time-varying $w(z)$ derived from the ECKS action—departs from this limit at late times, yielding $w_0 \approx -0.904$, $w_a \approx -0.153$, consistent with recent DESI BAO observations [1, 5].

As the Universe expands, the absolute densities of baryonic matter and topological dark matter [2] dilute as a^{-3}. Consequently, their fractional sum (Ω_m) decreases. Because the sum of all fractional densities must strictly remain 1 to satisfy the TIP (Eq. 1), the fractional weight of the frozen residual torsion (Ω_τ) is *mechanically* forced to increase over cosmic time to compensate for matter dilution (see Figure 1).

What the standard ΛCDM model interprets as a mysterious, dominant Dark Energy fluid ($\Omega_\Lambda \approx 0.7$) driving accelerated expansion is, in the ECF framework, simply the geometric response of spacetime taking up a larger fraction of the cosmic budget to prevent the Universe from curving upon matter dilution.

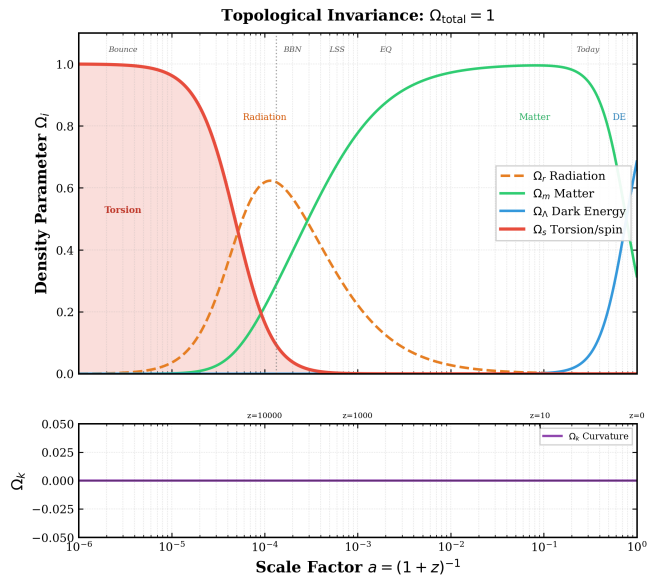

FIG. 1. **Topological Invariance.** Temporal evolution of fractional energy densities (Ω_i) in the ECF model. The bottom solid line ($\Omega_{total} = 1$) is the visual expression of the *Topological Invariance Principle*. The late-time emergence of the torsion density (Ω_τ, conventionally termed Dark Energy) acts mechanically as the geometric restoring force required to compensate for matter dilution and preserve absolute spatial flatness.

CONCLUSION

By elevating spatial flatness from a coincidental parameter to a fundamental gauge law, the ECF model eliminates the need for an exotic Dark Energy substance. The Topological Invariance Principle—$\Omega_{total} \equiv 1$ as an absolute geometric invariant—serves as the foundational postulate of the entire Einstein-Cartan trilogy: it constrains the spin-torsion sector in Foundation I, seeds the geometric dark sector in Foundation II, and governs the dynamics of cosmic voids and large-scale structure therein. Dark Energy is redefined as the dynamic fractional torsion required to maintain topological equilibrium against the dilution of matter. At leading order the frozen torsion mimics Λ, while the full geometric $w(z)$ provides a falsifiable deviation consistent with DESI data [5]. The comprehensive numerical proofs—the H_0 and S_8 tension resolution (Foundation I [1]) and the macroscopic modeling of topological Dark Matter via torsion knots (Foundation II [2])—are detailed in the companion regular articles. A direct observational consequence of the frozen torsion debt is a cosmic birefringence angle $\beta_{\mathrm{ECF}} = 0.35°$, testable by LiteBIRD [1].

[1] P. Fichant, "Foundation I: The Metric Universe — Extended Version. A Geometric Cosmology with Regular Bounce, Stiff-Era Expansion, and Testable Late-Time Signatures," Zenodo (2026), doi:10.5281/zenodo.19577447, `https://github.com/pfichant/spin-torsion-cosmology`.

[2] P. Fichant, "Foundation II: The Chiral Universe and the Three Fossils of the Bounce — Investigating Baryogenesis, the Geometric Dark Sector, and Cosmic Alignment within an Einstein-Cartan Framework," In preparation (2026).

[3] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020).

[4] N. J. Popławski, "Cosmology with torsion: An alternative to cosmic inflation," Phys. Lett. B **694**, 181 (2010).

[5] A. G. Adame *et al.* (DESI Collaboration), "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations," arXiv:2404.03002 (2024).